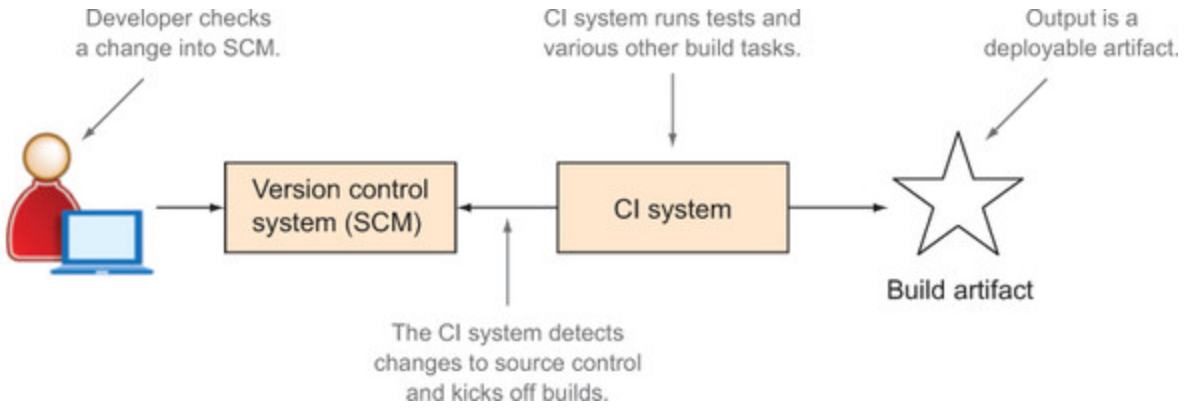
In today's digital world consumers expect the software they interact with every day to continuously improve. Mobile devices and web interfaces are ubiquitous and are evolving so rapidly that the average consumer of data expects interfaces to continually be updated and improved. To be able to provide your consumers the most competitive products, the development world has adapted by designing deployment systems that continuously integrate, test, and deploy changes. When used to their full potential, continuous practices allow development teams to hone their consumer's experience multiple times a day.

5.1.1. Continuous integration

Continuous development starts with CI, the practice of continuously building and testing your code as multiple team members update it. The simplest possible CI pipeline is shown in figure 5.3.

Figure 5.3. The simplest possible CI pipeline. When developers check in changes, the CI system builds an artifact.



In theory, CI allows teams to collaborate on the same codebase without stepping on each other's toes. SCM provides the source of truth for your codebase; multiple developers can work on the same software product at the same time and be aware of each other's changes. The CI system takes one or more of the changes that are made in SCM and runs a build script that at its simplest will ensure that the code you're working on compiles. But CI systems are basically servers that are designed to run multiple jobs as defined in the build script for an application and therefore are capable of